LESSON PLAN

U/W WEAPON (38)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Precision Measuring Tools – Introduction to Vernier Caliper, Screw Gauge and Feeler Gauge, Usage of common/ Precision Measuring tools.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Precision Measuring Tools – Introduction to Vernier Caliper, Screw Gauge and Feeler Gauge, Usage of common/ Precision Measuring tools.
5. Specific Objectives	38.1.1.1 Trainee will be able to explain principles of Correct Aircraft Maintenance Practices, tools and their usage

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Vernier Caliper, Screw Gauge, and Feeler Gauge usage; Safety precautions regarding tool handling and zero error correction; Instructor supervision during measurement exercises.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of precision measuring tools in aircraft maintenance?
(b) Name three common precision measuring tools used in engineering.
(c) Why is correct aircraft maintenance practice essential for tool usage?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Precision Measuring Tools • Principles of Correct Aircraft Maintenance Practices • Importance of Accuracy in Measurements	L & Ds	PPT, IB	15 min
(b)	• Introduction to Vernier Caliper • Introduction to Screw Gauge • Introduction to Feeler Gauge	L & Ds	PPT, PDF	20 min
(c)	• Usage of Common Precision Measuring Tools • Zero Error Correction Techniques • Maintenance and Storage of Tools	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the principle of operation of a Vernier Caliper.
(b) Describe the procedure for correcting zero error in a Screw Gauge.

12. Home Assignment and Follow Up	Prepare a brief report on the types of wires used in wire locking and their applications, to be discussed in LP 08.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Wire Locking – Introduction, Types of wires used in different wire locking and Wire locking procedure.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min